Smart Mood & Environment Control System

Project Banner

A smart Arduino-based system that monitors environmental conditions and responds accordingly. The system uses a DHT11 sensor to read temperature and humidity, displays the current mood on an OLED display, and triggers alerts or controls devices like fans or heaters via a relay module.

GitHub Stats

📌Features

- Real-time temperature and humidity monitoring using DHT11
- OLED display with animated mood face based on temperature
- LED and buzzer alerts when thresholds are exceeded
- Relay-controlled DC motor for environmental control (fan/heater)
- Customizable temperature and humidity thresholds
- Compact and modular design for IoT applications

🛠️Components Used

Component	Quantity	Purpose
Arduino UNO	1	Main controller
DHT11 Sensor	1	Temperature & Humidity sensing
OLED Display (0.96"/1.3" I^2C)	1	Display readings & animated mood face
Active Buzzer	1	Audio alert
LED + 220Ω resistor	1	Visual alert
Relay Module	1	Control DC Motor
DC Motor	1	Fan or actuator
Push Button	1	Optional manual control
Jumper Wires & Breadboard	-	Circuit connections

📖How It Works

1. Arduino reads temperature and humidity from DHT11.

2. OLED displays the values along with an animated mood face (Cold, Comfort, Hot, Extreme).
3. Temperature thresholds:
4. **High (>32°C):** LED ON, Buzzer ON, Relay activates Motor/Fan
5. **Low (<22°C):** LED blink, Buzzer ON, Relay OFF
6. User can press the button for optional control.

🧮Wiring Diagram

Wiring Diagram

Connections: - **DHT11:** VCC → 5V, GND → GND, DATA → D2 - **OLED:** VCC → 3.3V, GND → GND, SDA → A4, SCL → A5 - **LED:** Digital Pin 3 → 220Ω resistor → LED → GND - **Buzzer:** Digital Pin 4 → Buzzer → GND - **Button:** Digital Pin 5 → Button → GND - **Relay:** IN → D6, VCC → 5V, GND → GND, COM/NO → Motor

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💻Arduino Code Example

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <DHT.h>

#define DHTPIN 2
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);

#define LED_PIN 3
#define BUZZER_PIN 4
#define BUTTON_PIN 5
#define RELAY_PIN 6

#define TEMP_THRESHOLD_HIGH 32
#define TEMP_THRESHOLD_LOW 22

Adafruit_SSD1306 display(128, 64, &Wire, -1);

void setup() {
  pinMode(LED_PIN, OUTPUT);
  pinMode(BUZZER_PIN, OUTPUT);
  pinMode(BUTTON_PIN, INPUT);
  pinMode(RELAY_PIN, OUTPUT);

  Serial.begin(9600);
  dht.begin();
```

```
  if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
    Serial.println(F("SSD1306 allocation failed"));
    for (;;);
  }
  display.display();
  delay(2000);
}

void drawMoodFace(float temp) {
  display.drawCircle(64, 32, 20, SSD1306_WHITE);
  display.fillCircle(55, 25, 3, SSD1306_WHITE);
  display.fillCircle(73, 25, 3, SSD1306_WHITE);
  if (temp < 22) {
    display.drawLine(55, 43, 73, 43, SSD1306_WHITE); // Cold face
  } else if (temp <= 28) {
    display.drawLine(55, 40, 58, 43, SSD1306_WHITE);
    display.drawLine(58, 43, 70, 43, SSD1306_WHITE);
    display.drawLine(70, 43, 73, 40, SSD1306_WHITE); // Comfort smile
  } else if (temp <= 32) {
    display.drawLine(55, 43, 73, 40, SSD1306_WHITE); // Hot face
  } else {
    display.drawLine(60, 38, 60, 48, SSD1306_WHITE);
    display.drawLine(68, 38, 68, 48, SSD1306_WHITE); // Extreme shocked
  }
}

void loop() {
  float temp = dht.readTemperature();
  float hum = dht.readHumidity();
  if (isnan(temp) || isnan(hum)) return;

  display.clearDisplay();
  display.setTextSize(1);
  display.setTextColor(SSD1306_WHITE);
  display.setCursor(0,0);
  display.print(F("Temp: ")); display.print(temp); display.println(" C");
  display.print(F("Humidity: ")); display.print(hum); display.println(" %");

  drawMoodFace(temp);

  if (temp > TEMP_THRESHOLD_HIGH) {
    digitalWrite(LED_PIN, HIGH);
    digitalWrite(BUZZER_PIN, HIGH);
    digitalWrite(RELAY_PIN, HIGH);
  } else if (temp < TEMP_THRESHOLD_LOW) {
    digitalWrite(LED_PIN, LOW);
    digitalWrite(BUZZER_PIN, LOW);
```

```
    digitalWrite(RELAY_PIN, LOW);
  }

  display.display();
  delay(2000);
}
```

📏Screenshots

OLED Display preview, animated mood faces (replace with your own images)

OLED Screenshot

🔗Benefits

- Home or lab environmental monitoring
- Automated fan/heater control
- Animated mood display for fun visualization
- Easy to extend for IoT integration

📚References

- Adafruit SSD1306 OLED Library
- DHT Sensor Library
- Arduino Official Documentation
- GitHub Readme Stats